Air Quality

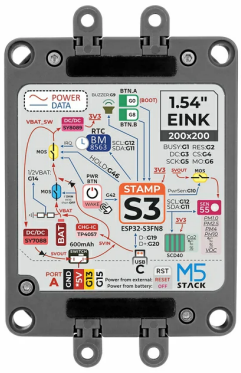
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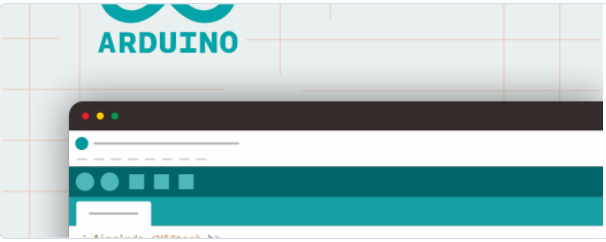


Description

Air Quality is a low-power, all-in-one air-quality monitoring device. It integrates the multifunctional air-quality sensor **SEN55** and the CO_2 sensor **SCD40**, enabling the measurement of PM1.0, PM2.5, PM4, PM10 particulate matter, temperature, humidity, VOC, and CO_2 concentration. Powered by the StampS3 MCU with 8MB Flash, it features a 1.54-inch E-Ink display (200 × 200) that retains the last image even when power is off. A built-in 600 mAh battery together with an RTC low-power management circuit supports sleep and timer wake-up functions.

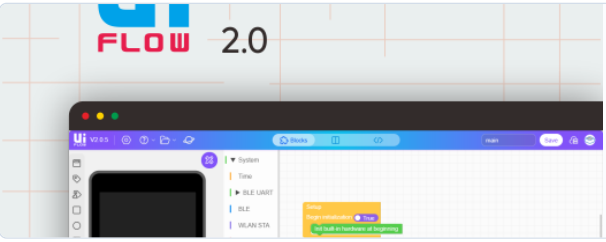
The factory firmware uploads data to M5’ s EZDATA cloud platform and automatically generates a monitoring page, allowing convenient remote access for users. The bottom of the device provides LEGO mounting holes, magnets, and four detachable mounting ears for versatile installation. It is ideal for long-term online monitoring in environments such as homes, schools, industrial sites, and medical facilities.

Tutorial



Arduino IDE

This tutorial shows how to control the Air Quality device using the Arduino IDE.



UiFlow2

This tutorial demonstrates how to control the Air Quality device with the UiFlow2 graphical programming platform.

Features

- SEN55 and SCD40 sensors
- 1.54-inch E-Ink display (200 × 200 px)
- Built-in 600 mAh battery
- Grove interface
- EZDATA cloud access
- RTC timer wake-up
- Development Platform
  - UiFlow2
  - Arduino IDE
  - ESP-IDF
  - PlatformIO

Includes

- 1 x Air Quality
- 1 x User Manual

Applications

- Home environment monitoring
- Industrial automation
- Medical facilities
- Research laboratories
- Remote monitoring
- HVAC system optimization
- Construction sites

Specifications

Specification	Parameter
Module Model	StampS3
SoC	ESP32-S3FN8 @ Xtensa® 32-bit LX7 dual-core processor, clock frequency 240MHz
USB	USB OTG, USB Serial/JTAG
Flash	8MB
Wi-Fi	2.4 GHz Wi-Fi
Display	GDEY0154D67 @ 1.54"
Resolution	200 × 200 px
SEN55	I2C address: 0x69
SCD40	I2C address: 0x62
Environmental Measurements	PM1.0, PM2.5, PM4, PM10, temperature, humidity, VOC, CO_2 concentration
RTC	RTC8563
Battery	600 mAh @ 3.7 V
Buttons	Button A (G0), Button B (G8), power button, reset/power-off
Grove Interface	HY2.0-4P
Buzzer	On-board passive buzzer
Mounting	LEGO holes, magnet, 4 x M3 detachable mounting ears
Operating Temperature	0 ~ 40 °C
Product Size	72.0 x 56.0 x 24.1 mm
Product Weight	91.9 g
Package Size	100.8 x 74.3 x 30.5 mm
Gross Weight	120.0 g

Learn

Power On/Off

- Power On: Press the “WAKE” button or wake up via the RTC timer IRQ signal. After wake-up, the program must set the HOLD pin (G46) to High (1) in initialization to keep the power on; otherwise, the device will return to sleep.



- Power Off: When no external USB power is connected, press the RST button, or set HOLD (GPIO46) = 0 in code to shut down.

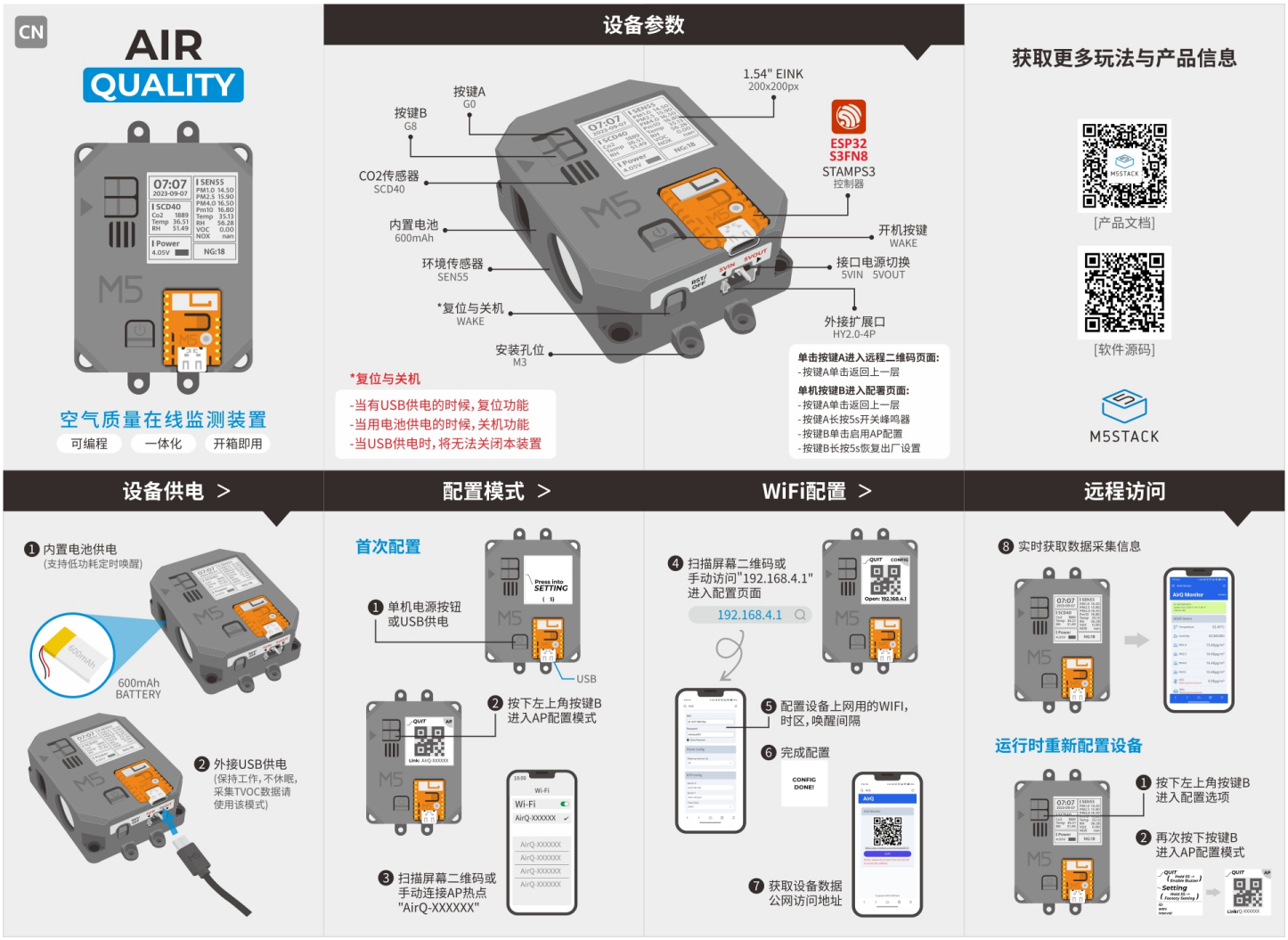


Download Mode

To enter download mode, first power off the device. While holding the BooT button on the StampS3 (or the G0 button on Air Quality), connect USB power. Release the button after power is applied.

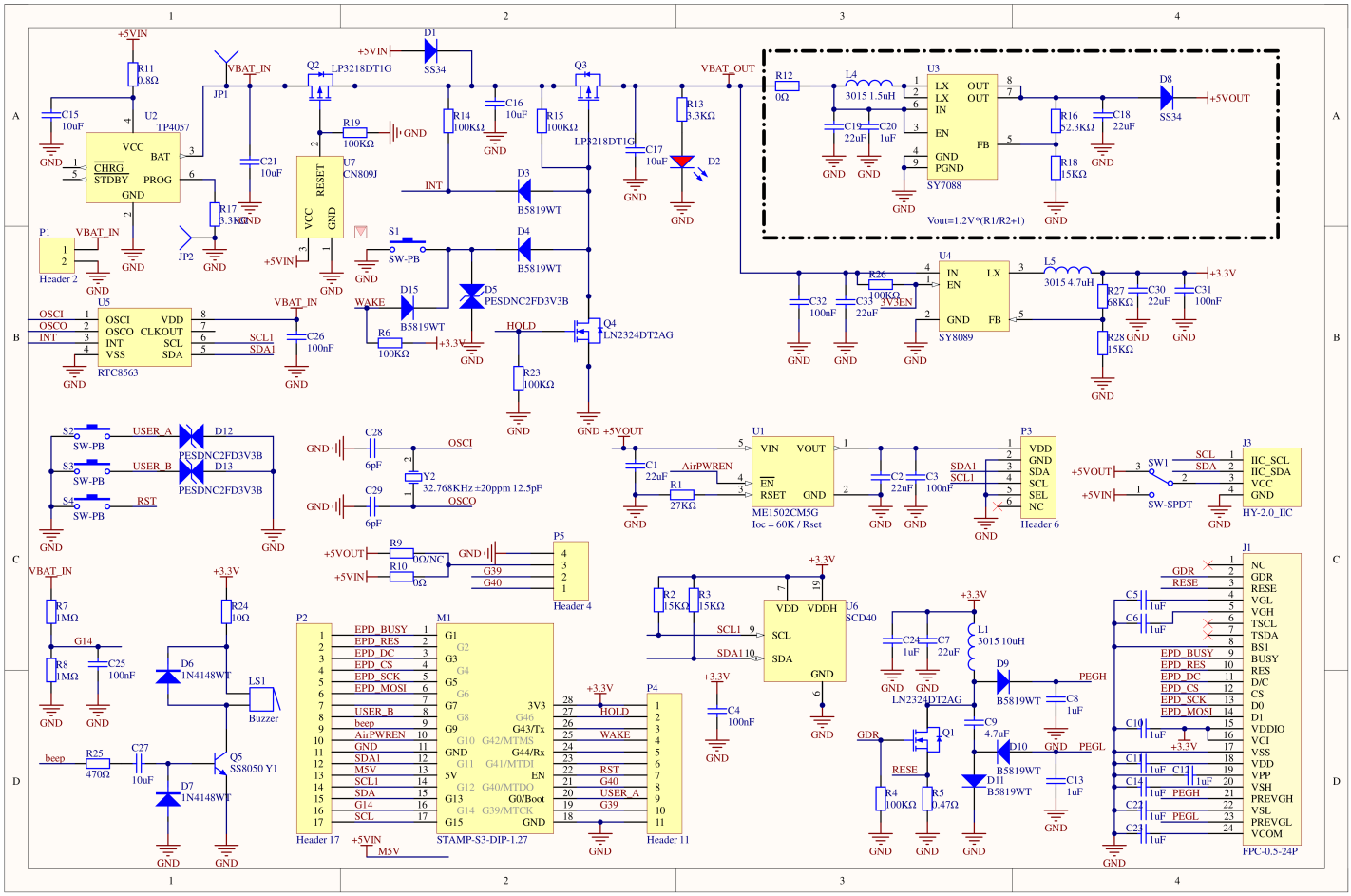


User Manual



Schematics

- Air Quality schematic PDF



PinMap

Power Control

ESP32-S3	G10	G46	G42	G14
SEN55 POWER SWITCH	AirPWREN			
HOLD		HOLD		
WAKE			WAKE	
BATTERY DETECT				G14

Display

ESP32-S3	G1	G2	G3	G4	G5	G6
GDEY0154D67	BUSY	RST	D/C	CS	SCK	MOSI

Input Interaction

ESP32-S3	G9	G0	G8
BEEP	beep		
BUTTON A		USER_A	
BUTTON B			USER_B

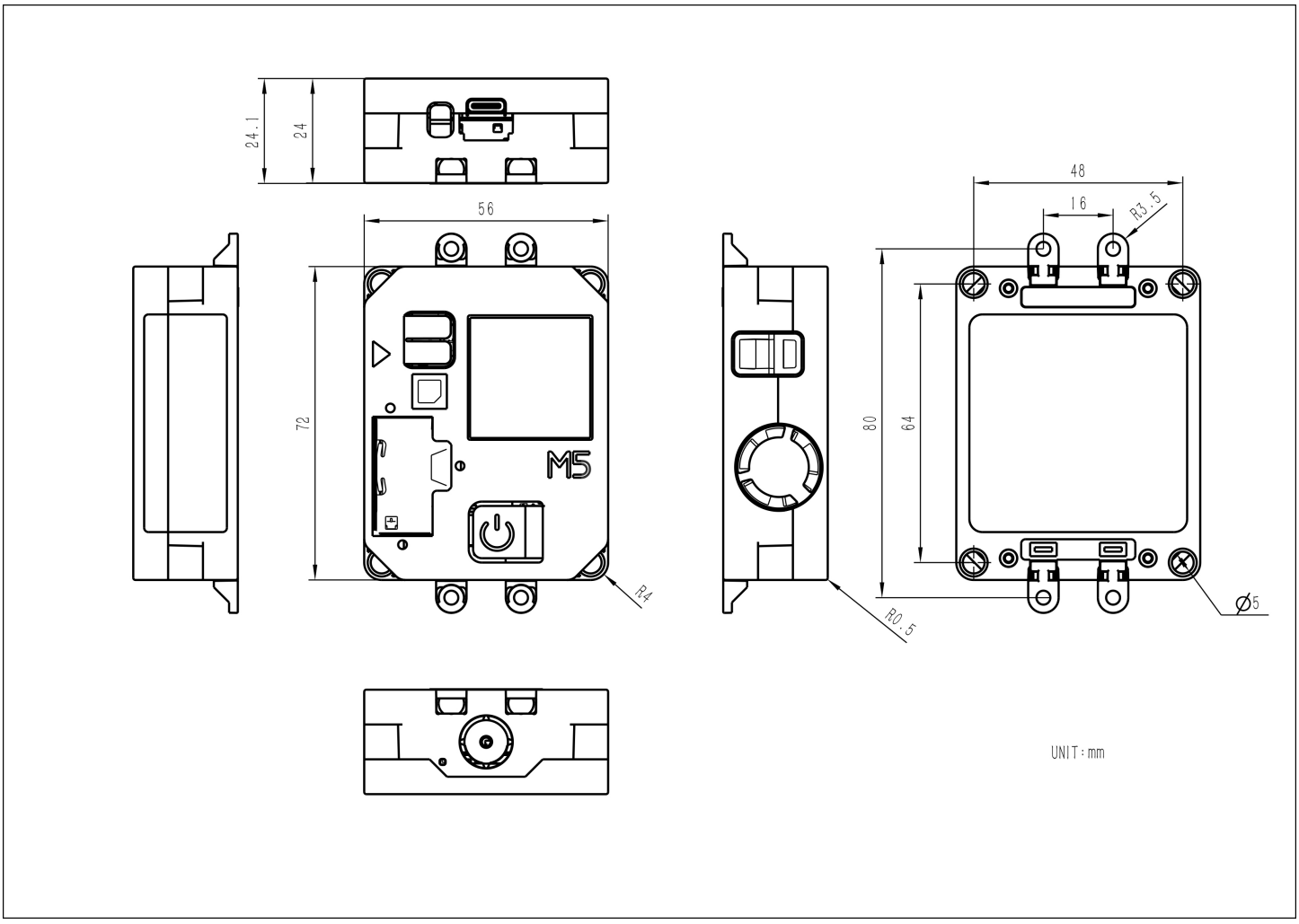
Sensors

ESP32-S3	G11	G12
SEN55	SDA	SCL
SCD40	SDA	SCL
RTC8563	SDA	SCL

HY2.0-4P

HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5 V	G13	G15

Model Size



Structure

- Air Quality Structure Files

Datasheets

- SEN55 Datasheet
- SCD40 Datasheet

- GDEY0154D67 Datasheet

Softwares

Arduino

- Air Quality Arduino Quick Start
- Air Quality Arduino M5Unified driver library
- Air Quality Arduino M5GFX driver library

UiFlow2

- Air Quality UiFlow2 Quick Start

PlatformIO

- Air Quality Factory Firmware

> **Restore to Factory State**
>
> If the Air Quality device has previously flashed UiFlow firmware and been bound to a user account, please unbind the device in the UiFlow2 device list before flashing the factory firmware. Otherwise, data may fail to upload to EzData after reflashing.

```
[env:m5stack-stamp-s3]
platform = espressif32
board = esp32-s3-devkitc-1
framework = arduino
upload_speed = 1500000
build_flags =
    -DESP32S3
    -DCORE_DEBUG_LEVEL=5
    -DARDUINO_USB_CDC_ON_BOOT=1
    -DARDUINO_USB_MODE=1
lib_deps =
    M5Unified=https://github.com/m5stack/M5Unified
```

Easyloader

Easyloader	Download	Note
Air Quality User Demo	download	/

Other

- Return Air Quality to Factory Firmware

Video